SUPPORTING ADULT GUIDANCE

Bounds of a rounded value

Percentages, Ratio, Growth and Bounds · Lesson 8. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The lesson opens with a crowd of 40 000 to the nearest thousand: learners test five possible real crowds. The main activity, the Rounding Zoom, is in Connection. Learners choose a degree of accuracy (nearest 10, nearest whole number, 1 or 2 decimal places, 1 or 2 significant figures) and a rounded value. A probe moves along a number line in exact steps and says what each value rounds to, so learners find the edges themselves before pressing Show the interval. The interval is then shaded with a filled dot at the lower bound and an open dot at the upper bound, and the readout gives the unit, half a unit, both bounds and the error interval. Zoom out shows the neighbouring intervals tiling the line. Six questions, quick bounds, true-or-not statements, a stretch on why the bounds are half a unit away, and a second Rounding Zoom for the Creativity briefs follow.

Driving Question: *What could it have been before rounding?*

The mathematics, in plain terms

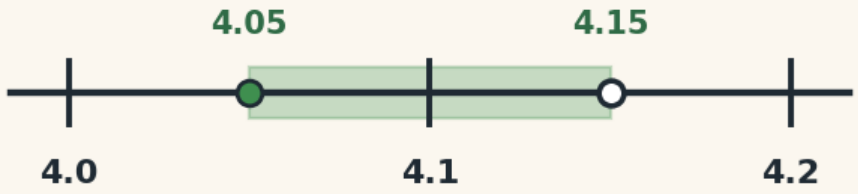
A value rounded to the nearest u as R lies in $[R - u/2, R + u/2)$ when halves round up. The upper bound is the supremum of the interval but not a member of it. Examinations accept the upper bound written as $R + u/2$ (for example 7.35), not as a recurring 9. For significant figures, the unit is the place value of the last significant figure.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — To 1 decimal place

A length is 4.1 m to 1 decimal place. Find its bounds and write its error interval.

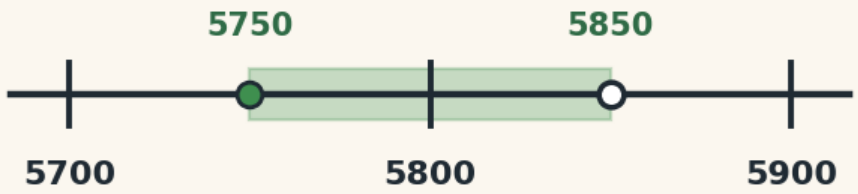


1. The unit	1 decimal place: the unit is 0.1
2. Half a unit	Half of 0.1 is 0.05
3. The bounds	4.1 – 0.05 = 4.05; 4.1 + 0.05 = 4.15
4. Check	4.05 rounds to 4.1; 4.15 rounds up to 4.2

$4.05 \leq x < 4.15$.

Example 2 — To 2 significant figures

A town's population is 5800 to 2 significant figures. Find its bounds.



1. The unit	The second significant figure (8) is in the hundreds: the unit is 100
2. Half a unit	Half of 100 is 50
3. The bounds	5800 – 50 = 5750; 5800 + 50 = 5850

$5750 \leq P < 5850$.

Probe first, then the rule

Learners find the edges by testing values and reading what they round to before the interval is shown. The half-unit rule is then something they have seen, and the open upper end comes from watching that value round up.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The crowd, the idea card, the Rounding Zoom (main activity), six questions.	~20 min
Movement	Quick bounds: eight in a row.	~6 min
Reflection	True or not; the half-unit stretch.	~8 min
Creativity	Make the interval: six briefs on a second Rounding Zoom.	~6 min
Rest	A good-enough picture, not the whole story.	~2 min
Nutrition	Practical: reading measurements honestly.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding or taking away a whole unit (6.3 and 6.5 for 6.4).
Ask: “Which values round to 6.4? Where do they stop?”

Writing the upper bound as 6.49 or 6.4999….
Ask: “What is halfway between 6.4 and 6.5?”

Using the same bounds whatever the accuracy.
Ask: “Rounded to what?”

Finding the wrong unit for significant figures.
Ask: “Which place is the last significant figure in?”

Using ≤ at both ends.
Ask: “Does the upper bound round to the value, or up?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: rounding to decimal places and significant figures. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says the bounds of 2300 to the nearest 100 are 2200 and 2400. The aim is a question that sends them to the halfway points.

- “What does 2280 round to? And 2240?”
- “Where do the values stop rounding to 2300?”

- “How far is that from 2300?”
- “So what are the bounds?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why is the upper bound written as 7.35 if it is not included?
It is the edge of the interval. The error interval uses < to show it is not included, which is the convention in examinations.

Does it matter that halves round up?
Yes: it is why the lower bound is included and the upper bound is not.

Is the stretch required?
No. The secure core is finding the unit, halving it and writing the error interval.

What comes next?
Lesson 9: calculating with bounds, choosing which bounds give the largest and smallest results.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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